

CLMS Product Filenaming - Design Principles

Copernicus Land Monitoring Service - Technical
Library



Author: **European Environment Agency (EEA)**

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Contact:

European Environment Agency (EEA)

Kongens Nytorv 6

1050 Copenhagen K

Denmark

<https://land.copernicus.eu/>

1. CLMS Product Filenaming - Design Principles

Grounded in the parsEO schema registry, CDSE product catalogue, and current filenaming conventions.

1.1 Purpose and Scope

These principles govern the filenaming of **all new CLMS product deliveries** - new products, reprocessings, and new versions of existing products. They describe the target convention; existing files on CDSE under legacy names remain as-is but are non-compliant.

1.2 Principle 1 - Filename Structure: fields, delimiter, extension

Rule: A CLMS filename consists of a stem (base name) and if required (e.g. not folder names) an extension, separated by a period .:

```
{stem}.{extension}
```

The stem is composed of **fields** - descriptive elements separated by underscore _. Each field carries a specific meaning:

```
field1_field2_field3_..._fieldN.ext
```

Fields may not be empty. A field delimiter may not appear at the beginning or end of a filename, nor twice in a row.

Extension: The file extension (.tif, .gpkg, .xml) indicates the file format and is not part of the naming convention.

Compliant: CLMS_VLCC-GRA_S2021_R10m_E27N48_3035_V01-R00.tif

Non-compliant: _CLMS_VLCC-GRA_... (leading delimiter), CLMS_VLCC-GRA__S2021_... (double delimiter)

1.3 Principle 2 - Allowed Characters: uppercase, digits, underscore, hyphen

Rule: Filenames MUST use only the following ASCII characters:

Character	ASCII	Usage
A-Z	65-90	Uppercase letters only (no lowercase)
0-9	48-57	Digits
_	95	Field delimiter
-	45	Within-field separator

The period . (ASCII 46) is reserved exclusively for the file extension separator. Spaces, lowercase letters, and all other special characters are NOT permitted.

Rationale: A restricted character set enables deterministic, case-insensitive parsing across all platforms. Uppercase-only avoids case-sensitivity issues between Linux (case-sensitive) and Windows/macOS (case-insensitive).

Compliant: CLMS, V01-R00, 054-0154-IW1, R20-5m, VLCC-GRA

Non-compliant: clms_EGMS, V01_r00, P20x5m, data file.tif

1.4 Principle 3 - Maximum Filename Length

Rule: The complete filename (stem + extension) MUST NOT exceed 255 characters. The fully qualified file name (including path) SHOULD NOT exceed 260 characters.

Recommendation: Keep filenames under 100 characters to safely accommodate both file name and storage path across all platforms.

Rationale: The 255-character limit is the smallest common denominator across Windows (260 char FQFN), macOS (1024), and Linux (4096). The 100-character recommendation preserves room for deep directory structures.

1.5 Principle 4 - Delimiter: Underscore _ between fields, hyphen - within fields

Rule: Use _ to separate all top-level fields. The hyphen - is the **within-field separator**: it joins the variable (and sub-variables) to the product code (VLCC-GRA, WSI-SP-SCD), joins version to revision (V01-R00), and appears where a notation itself requires it (e.g., 054-0154-IW1 for IW burst tile identifiers, C2018-2021 for change periods). It is never a cross-field delimiter.

Rationale: Consistent underscore delimiters make tokenisation deterministic across all product families. The hyphen carries all intra-field structure, so splitting on _ always yields the same field count within a product family.

Compliant:

- CLMS_VLCC-GRA_S2021_R10m_E27N48_3035_V01-R00.tif
- CLMS_EGMS-L2a-A_T20190101P5Y_R20-5m_054-0154-IW1_V01-R00.tif (hyphens inside EGMS-L2a-A and 054-0154-IW1 are semantic)
- CLMS_CLCPLUS-LCU_S2023_R10m_E48N37_3035_V01-R00.tif

Edge case: Hyphenated geographic names (e.g., "Bourg-en-Bresse") is a single token; the hyphen is part of the name, not a delimiter.

1.6 Principle 5 - Field Order: Variable hyphen-appended to the product code, token count invariant

Rule: Fields appear in a fixed order per product family. The variable is **not a standalone field**: it is hyphen-appended to the product code in position 2 ({CODE}-{VARIABLE}[-{SUBVARIABLE}]). Underscore-separated token count is invariant across a product family - a sub-product identifier is never a separate positional field; it is encoded within the code-variable compound:

Canonical template:

CLMS_{CODE}-{VARIABLE}[-{SUBVARIABLE}][_ {TEMPORAL}]{RES}_{EXTENT}_{EPSG}_{VERSION}-{REVISION}

Sub-products within a family (e.g., SP, SWS, WDS within WSI) extend the compound with a further hyphen (WSI-SP-SCD, WSI-WDS-SSC). This keeps the underscore token count identical for every file in the family.

Rationale: Code and variable together identify **what** the file is; the remaining fields identify **when, where, and which version**. Placing the full product identity at position 2 groups all files of a product and its layers together in directory listings, and a parser always knows which position holds which field - no branching logic based on which sub-product happens to be present.

Compliant:

- CLMS_VPP2-ST-PPI_A20240101P10D_R10m_T33UVS_V01-R00.tif
- CLMS_WSI-SP-SCD_A20200901P1Y_R20m_T38TKL_V01-R00.tif
- CLMS_WSI-WDS-SSC_20210217T053159_R60m_T32TNS_S1B_V01-R00.tif (sensing datetime, sub-product in the code-variable compound)

1.7 Principle 6 - Programme Prefix: CLMS_ mandatory, no exceptions

Rule: Every CLMS product filename MUST begin with CLMS_. No exceptions. No product is exempt. This applies to all new deliveries, reprocessings, and new product versions across the entire CLMS portfolio (EEA, JRC, contractors).

Rationale: The prefix makes programme identity parseable from the filename alone. Any file without CLMS_ is immediately identifiable as non-conformant.

Compliant:

- CLMS_VLCC-GRA_S2021_R10m_E27N48_3035_V01-R00.tif
- CLMS_VPP2-ST-PPI_A20240101P10D_R10m_T33UVS_V01-R00.tif
- CLMS_UA-LCU_S2021_V025ha_DK004L3_3035_V01-R00.gpkg
- CLMS_CLCPLUS-LCU_S2018_R10m_EUROPE_3035_V01-R00.tif
- CLMS_CLC-LCU_S2018_V25ha_EUROPE_3035_V01-R00.gpkg
- CLMS_EGMS-L3-U_T20190101P5Y_R100m_E28N49_3035_V01-R00.tif

1.8 Principle 7 - Product Code: Short, uppercase, registered

Rule: Product codes must be registered in the CLMS product code registry before schema authoring. Uppercase alphanumeric only. Prefer 4-7 characters. Retain the HR prefix only when it is part of the established acronym.

Compliant: VLCC, VPP2, WSI, EUHYDRO, CLCPLUS, UA

Non-compliant: HRVPP2 (redundant HR), VPP-2 (hyphen in code), any unregistered code

1.9 Principle 8 - Temporal Coverage: Format matches product type

Rule:

Situation	Format	Examples	Definition
Sensing instant	{YYYYMMDD}T{HHMMSS}	20210217T053159	Single acquisition at a specific date+time (e.g. satellite

Situation	Format	Examples	Definition
			scene). Zero-duration temporal instant. Uses bare ISO 8601 <code>dateTime</code> lexical form - the <code>T</code> separator and fixed-width format make it self-identifying, so no semantic prefix is needed.
Aggregated	A{YYYYMMDD}P{duration}	A20240101P10D, A20240101P1M, A20240101P1Y	A 2D snapshot where pixel values are derived by compositing, averaging, or selecting from multiple observations over time using a defined algorithm. The temporal dimension is collapsed to a single representative value per pixel; the file is not designed for temporal querying.
Status	S{YYYY}	S2021	A static thematic map or interpreted classification representing a single reference year. The product does not represent a statistical aggregate of repeated measurements; it is the result of a single survey, interpretation, or classification exercise.
Change	C{YYYY}-{YYYY}	C2018-2021	A map of transitions or differences between two Status reference years. The two years are the start and end of the change period. The output is a 2D change layer (not a time-series stack).
Timeseries	T{YYYYMMDD}P{duration}	T20190101P5Y	A single file containing preserved multi-temporal measurements where the temporal dimension is queryable (e.g. as internal bands, layers, or a time-enabled data cube). Designed for temporal analysis, not as a collapsed 2D snapshot.
Forecast	F{YYYYMMDD}P{duration}	(Reserved)	Reserved/Proposed. For future forecast/modelled products. Not currently in use.

Temporal reference standard: All temporal formats follow **ISO 8601-1:2019** (Date and time - Representations for information interchange). The `P{duration}` suffix uses the ISO 8601 duration format (`P10D`, `P1M`, `P1Y`). Reference: <https://www.iso.org/standard/70907.html>

Edge case: Multi-seasonal products encode the season in the code-variable compound (`-S1`, `-S2`), not in the temporal field. This is consistent with external standards (CF Conventions, NASA EOSDIS) which treat season as a data attribute, not a temporal primitive.

Compliant examples:

- `CLMS_VPP2-ST-PPI_A20240101P10D_...` (aggregated, dekadal)
- `CLMS_VLCC-GRA_S2021_...` (annual status)
- `CLMS_UA-LCU_C2018-2021_...` (change between two Status years)
- `CLMS_WSI-WDS-SSC_20210217T053159_R60m_T32TNS_S1B_V01-R00.tif` (sensing datetime)
- `CLMS_EGMS-L3-U_T20190101P5Y_R100m_E28N49_3035_V01-R00.tif` (timeseries, multi-year)

1.10 Principle 9 - Resolution: `R{value}[unit]` for raster, `V{value}[unit]` for vector and point grid

Rule: Raster and gridded products use `R{value}[unit]` with no zero-padding: `R10m`, `R1km`, `R100m`. The value is the integer pixel/grid spacing in metres. Irregular grid spacings join both values with a hyphen within the field: `R20-5m` (e.g., EGMS L2 20×5 m posting).

The `V` prefix covers two cases:

- **Vector minimum mapping units** - `V{value}ha` (e.g., `V025ha` for Urban Atlas 0.25 ha, `V25ha` for CLC 25 ha). Sub-1 ha values use a leading zero with no decimal point.
- **Point grid posting spacing** - `V{value}m` for regular grids (e.g., `V100m` for EGMS L3) or `V{value}-{value}m` for irregular grids (e.g., `V20-5m` for EGMS L2 point data)

Only `R` and `V` exist - there is no separate point-grid prefix.

Compliant: `R10m`, `R1km`, `R100m`, `R20-5m`, `V025ha`, `V25ha`, `V100m`, `V20-5m`

Edge case: Products with no meaningful spatial resolution (e.g., nation-level statistical vector layer) omit the resolution token.

1.11 Principle 10 - Spatial Extent: Identifies the area covered by this specific file

Rule: This field identifies the spatial extent of the individual file - not the overall product coverage. It can be a grid tile, a satellite scene, an administrative boundary, or a continental code:

Extent type	Description	Examples
MGRS	Sentinel-2 raster tile	T33UVS
LAEA 100km	Pan-European composite on INSPIRE/ ETRS89-LAEA grid	E27N48
IW burst	Sentinel-1 swath (burst-level)	054-0154-IW1
FUA code	Functional Urban Area, city-level vector	DK004L3
Country code	Nation-level vector	DE, IT
Delivery Unit	Tiling system for large-area products	DU001

Extent type	Description	Examples
Continental	Full European continent coverage	EUROPE
Hemispherical	Global product subset for one hemisphere	NORTHERNHEMISPHERE, SOUTHERNHEMISPHERE
Global	Full global coverage	GLOBAL

Spatial extent identifiers are _-separated from adjacent tokens. Never fuse them with EPSG or resolution.

Note: This field describes the extent of the **file**, not the product. A pan-European product may be delivered as individual MGRS tiles, each with a different spatial extent. Likewise, **EUROPE** means this specific file covers the entire continent - it does not mean the product is “pan-European” by design.

Compliant:

- CLMS_VPP2-ST-PPI_..._T33UVS_... (MGRS tile)
- CLMS_VLCC-GRA_S2021_R10m_E27N48_... (LAEA tile)
- CLMS_CLCPLUS-LCU_S2023_R10m_E48N37_... (LAEA tile)
- CLMS_UA-LCU_S2021_V025ha_DK004L3_... (FUA code)
- CLMS_EGMS-L2a-A_..._054-0154-IW1_... (IW burst)
- CLMS_EUHYDRO-NET-LINE-EUROPE_... (continental coverage)
- CLMS_WSI-WDS-SSC_20210217T053159_R60m_T32TNS_S1B_V01-R00.tif (MGRS, SAR sensor)

1.12 Principle 11 - EPSG Code: Bare code, required for projected CRS, omit when CRS is implicit

Rule: Standalone _-separated token containing the bare EPSG code with **no zero-padding** (3035). Required for products with a projected CRS (e.g., LAEA 100km extents). Omit when the CRS is implicit in the spatial extent system (e.g., MGRS tiles imply UTM, so EPSG is redundant). Also omit for CRS-independent files (metadata XML, archive containers).

Rationale: The bare code is the EPSG identifier exactly as registered (<https://epsg.org>) - 3035, 4326, 32633 - so the token matches the authority and needs no transformation for lookups. EPSG codes are 4–5 digits; the field is bounded, and the surrounding fixed field order keeps parsing deterministic without artificial padding.

Compliant:

- CLMS_VLCC-GRA_S2021_R10m_E27N48_3035_V01-R00.tif (LAEA tile, EPSG 3035 required)
- CLMS_CLCPLUS-LCU_S2023_R10m_E48N37_3035_V01-R00.tif
- CLMS_VPP2-ST-PPI_A20240101P10D_R10m_T33UVS_V01-R00.tif (MGRS tile - EPSG omitted, CRS implicit)

Non-compliant: 03035 (zero-padded), EPSG3035 (prefixed)

1.13 Principle 12 - Version: V{XX}-R{YY} single hyphenated field

Rule: A single compound token: V{XX} (major, 2-digit zero-padded), hyphen, R{YY} (revision, 2-digit zero-padded). Never use packed V{XXX}.

Compliant: V01-R00, V01-R01, V02-R00

1.14 Principle 13 - Variable: Hyphenated composite appended to the product code

Rule: The variable identifies the measured or mapped quantity. It is hyphen-appended to the product code at position 2, forming the product identity compound:

```
{CODE}-{PRODUCT_TYPE}[-{PARAMETER}][-{SEASON}][-QFLAG]
```

- **PRODUCT_TYPE:** layer type code (ST, VPP, GPP, SP, LCU)
- **PARAMETER:** specific measurement (PPI, SOSD, TOTAL, SCD)
- **SEASON:** S1 or S2 for multi-seasonal products
- **QFLAG:** literal suffix for quality flag layers - always last

LCU (Land Cover / Land Use) is the shared variable for all status and change land cover/use products (UA, CLC, CLCPLUS, RZ, ...); status vs change is expressed by the temporal field (S{YYYY} vs C{YYYY}-{YYYY}), not by the variable.

Compliant:

VPP2-ST-PPI	→ VPP2-ST-PPI-QFLAG
VPP2-VPP-SOSD-S1	→ VPP2-VPP-SOSD-S1-QFLAG
VPP2-GPP-TOTAL	→ VPP2-GPP-TOTAL-QFLAG
WSI-SP-SCD	→ WSI-SP-SCD-QFLAG
UA-LCU / CLC-LCU / CLCPLUS-LCU	

1.15 Principle 14 - Production Date: Metadata-only unless operationally required

Rule: Omit production date from filenames. Include it only when multiple production runs for the same temporal period coexist in the same distribution channel. When present: {YYYYMMDD} as the last token before the extension.

Compliant: CLMS_EUHYDRO-NET-LINE_EUROPE_3035_V02-R00_20261015.gpkg

1.16 Principle 15 - Sensor Token: Omit unless multi-sensor discrimination is required

Rule: No standalone sensor token in the filename. The sensor is implicit in the product code. Include a sensor token only when the product supports multiple sensor sources distinguishable at file level (e.g., SAR platform identity _S1B_).

Compliant:

- CLMS_WSI-SP-SCD_A20200901P1Y_R20m_T38TKL_V01-R00.tif
- CLMS_WSI-WDS-SSC_20210217T053159_R60m_T32TNS_S1B_V01-R00.tif (S1B sensor token required for SAR discrimination)

1.17 Principle 16 - Quality Layers: Separate file, hierarchical suffix -QFLAG

Rule: Quality layers are separate files. Same stem as the data layer, with -QFLAG appended to the appropriate hierarchical level of the code-variable compound. No other field may differ between a data file and its quality file.

The -QFLAG suffix attaches at the level it applies to:

- VPP2-VPP-SOSD-QFLAG - quality applies to the VPP-SOSD product only
- VPP2-VPP-QFLAG - quality applies to all VPP sub-products (SOSD, TOTAL, SEASONAL, etc.)
- VPP2-ST-PPI-QFLAG - quality applies to the PPI measurement only

Compliant:

```
CLMS_VPP2-ST-PPI_A20240101P10D_R10m_T33UVS_V01-R00.tif ← data
CLMS_VPP2-ST-PPI-QFLAG_A20240101P10D_R10m_T33UVS_V01-R00.tif ← quality
```

1.18 Principle 17 - Schema Registration: parseEO schema BEFORE CDSE onboarding

Rule: Every new CLMS product must have a registered product code, a parseEO schema with `status: current`, and a naming crosswalk before the first CDSE delivery. The schema defines the filename; CDSE filenames are not reverse-engineered into schemas after the fact.

1.19 What Cannot Be Standardised

Some aspects genuinely differ between product families and should not be forced into a single mould:

1. **Raster vs. vector/point-grid resolution** - R10m and V25ha are different physical quantities; V covers both vector MMU (V{value}ha) and point grids (V{value}m)
2. **Pan-European vs. local tile encoding** - LAEA grid tiles (E27N48) for pan-European products, FUA codes (DK004L3) for city-level vector - these are incompatible spatial encodings, both valid
3. **Annual status vs. scene-level temporal format** - S{YYYY} for static maps, {YYYYMMDD}T{HHMMSS} for single acquisitions - the right format depends on the product's temporal nature
4. **Multi-resolution families** - the set of valid resolution values varies per product; each schema enumerates its own
5. **Production date inclusion** - reflects operational policy; included when corrections under the same version are expected

1.20 Principles Summary

#	Principle	Rule
1	Filename structure	Fields separated by _, stem.extension
2	Allowed characters	A-Z, 0-9, _, - only; no lowercase, no spaces
3	Max filename length	≤255 chars stem+extension, recommend ≤100
4	Delimiter	_ between fields; - within fields (code-variable compound, version-revision, notations)

#	Principle	Rule
5	Field order	Variable hyphen-appended to product code at position 2; canonical template defines field sequence
6	Prefix	CLMS_ mandatory for all products, no exceptions
7	Product code	Short, uppercase, registered in code registry
8	Temporal	A{start}P{duration} (aggregated), S{YYYY} (status), C{YYYY}-{YYYY} (change), T{start}P{duration} (timeseries)
9	Resolution	R{XX}m (raster/grid), V{value}ha (vector MMU), V{value}m (point grid) - hyphen for irregular grids
10	Spatial extent	MGRS / LAEA / IW burst / FUA / country / DU / EUROPE - identifies the area of the specific file, not the product coverage
11	EPSG	Bare EPSG code, no zero-padding (3035); required for projected CRS, omit when CRS is implicit in the spatial extent
12	Version	V{XX}-R{YY} single hyphenated field; no packed V{XXX}
13	Variable	{CODE}-{TYPE}[-{PARAM}][-{SEASON}][-QFLAG] compound at position 2
14	Production date	Metadata-only unless operationally required
15	Sensor	Omit unless multi-sensor discrimination needed
16	Quality	Separate file, hierarchical -QFLAG suffix, identical stem
17	Registration	parsEO schema before CDSE onboarding

Document status: Draft for review. Grounded in the parsEO schema registry and VPP2 naming crosswalk.